

nano sample_weather.txt

```
2024-01-01 10 5 12
2024-01-02 12 6 10
2024-01-03 14 7 15
2024-01-04 16 8 20
2024-01-05 18 9 25
2024-01-06 20 10 18
2024-01-07 22 11 14
2024-01-08 24 12 16
2024-01-09 26 13 20
```

nano WeatherAverage.java

```
import java.io.IOException;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WeatherAverage {

    public static class WeatherMapper
        extends Mapper<Object, Text, Text, DoubleWritable> {

        private Text keyOut = new Text();
        private DoubleWritable valueOut = new DoubleWritable();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            String[] parts = value.toString().split(" ");

            double temp = Double.parseDouble(parts[1]);
            double dew = Double.parseDouble(parts[2]);
            double wind = Double.parseDouble(parts[3]);

            context.write(new Text("TEMP"), new DoubleWritable(temp));
            context.write(new Text("DEW"), new DoubleWritable(dew));
            context.write(new Text("WIND"), new DoubleWritable(wind));
        }
    }
}
```

```
}
```

```
public static class WeatherReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
        throws IOException, InterruptedException {

        double sum = 0;
        int count = 0;

        for (DoubleWritable val : values) {
            sum += val.get();
            count++;
        }

        double avg = sum / count;
        context.write(key, new DoubleWritable(avg));
    }
}
```

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "weather average");

    job.setJarByClass(WeatherAverage.class);

    job.setMapperClass(WeatherMapper.class);
    job.setReducerClass(WeatherReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ export HADOOP_CLASSPATH=$(hadoop
classpath)
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ javac -classpath "$HADOOP_CLASSPATH" -d .
WeatherAverage.java
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ jar -cvf WeatherAvg.jar *.class
added manifest
adding: WeatherAverage$WeatherMapper.class(in = 1866) (out= 784)(deflated 57%)
adding: WeatherAverage$WeatherReducer.class(in = 1719) (out= 729)(deflated 57%)
adding: WeatherAverage.class(in = 1498) (out= 808)(deflated 46%)
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ hdfs dfs -mkdir -p /weather/input
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ hdfs dfs -put sample_weather.txt
/weather/input
```

```
omkar@LAPTOP-S6G2S414:~/weather-job$ hadoop jar WeatherAvg.jar WeatherAverage
/weather/input /weather/output
2026-04-21 20:51:17,739 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to
ResourceManager at /0.0.0.0:8032
2026-04-21 20:51:18,240 WARN mapreduce.JobResourceUploader: Hadoop command-line
option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-04-21 20:51:18,268 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding
for path: /tmp/hadoop-yarn/staging/omkar/.staging/job_1776802751232_0004
2026-04-21 20:51:18,600 INFO input.FileInputFormat: Total input files to process : 1
2026-04-21 20:51:18,693 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-21 20:51:18,937 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_1776802751232_0004
2026-04-21 20:51:18,937 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-21 20:51:19,116 INFO conf.Configuration: resource-types.xml not found
2026-04-21 20:51:19,117 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-04-21 20:51:19,202 INFO impl.YarnClientImpl: Submitted application
application_1776802751232_0004
2026-04-21 20:51:19,248 INFO mapreduce.Job: The url to track the job: http://LAPTOP-
S6G2S414.:8088/proxy/application_1776802751232_0004/
2026-04-21 20:51:19,249 INFO mapreduce.Job: Running job: job_1776802751232_0004
2026-04-21 20:51:26,418 INFO mapreduce.Job: Job job_1776802751232_0004 running in
uber mode : false
2026-04-21 20:51:26,420 INFO mapreduce.Job: map 0% reduce 0%
2026-04-21 20:51:31,507 INFO mapreduce.Job: map 100% reduce 0%
2026-04-21 20:51:37,557 INFO mapreduce.Job: map 100% reduce 100%
2026-04-21 20:51:38,582 INFO mapreduce.Job: Job job_1776802751232_0004 completed
successfully
2026-04-21 20:51:38,726 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=3966
    FILE: Number of bytes written=559685
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=1906
```

HDFS: Number of bytes written=68
HDFS: Number of read operations=8
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

Job Counters

Launched map tasks=1
Launched reduce tasks=1
Data-local map tasks=1
Total time spent by all maps in occupied slots (ms)=3049
Total time spent by all reduces in occupied slots (ms)=3019
Total time spent by all map tasks (ms)=3049
Total time spent by all reduce tasks (ms)=3019
Total vcore-milliseconds taken by all map tasks=3049
Total vcore-milliseconds taken by all reduce tasks=3019
Total megabyte-milliseconds taken by all map tasks=3122176
Total megabyte-milliseconds taken by all reduce tasks=3091456

Map-Reduce Framework

Map input records=90
Map output records=270
Map output bytes=3420
Map output materialized bytes=3966
Input split bytes=119
Combine input records=0
Combine output records=0
Reduce input groups=3
Reduce shuffle bytes=3966
Reduce input records=270
Reduce output records=3
Spilled Records=540
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=54
CPU time spent (ms)=1610
Physical memory (bytes) snapshot=495706112
Virtual memory (bytes) snapshot=5508972544
Total committed heap usage (bytes)=365953024
Peak Map Physical memory (bytes)=270974976
Peak Map Virtual memory (bytes)=2753036288
Peak Reduce Physical memory (bytes)=224731136
Peak Reduce Virtual memory (bytes)=2755936256

Shuffle Errors

BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0

WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=1787
File Output Format Counters
Bytes Written=68

```
omkar@LAPTOP-S6G2S414:~/weather-job$ hdfs dfs -cat /weather/output/part-r-00000  
DEW  20.188888888888889  
TEMP 40.066666666666667  
WIND 35.155555555555556
```